

Lecture Study Guide Question Bank



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Syllabus Question 1.1

Set 0 A

If a student is getting low scores on learning assessments, according to the syllabus, which of the following is LEAST likely to be helpful in improving future scores?

- A. Increasing regular study and review of course material
- B. Testing oneself to develop one's own judgment of learning
- C. Obtaining feedback on which specific items were missed on learning assessments
- D. Increasing focus on big-picture concepts
- E. Increasing use of lecture videos

Note: Question set 0A includes questions from sections of the syllabus. After this section, the remainder of the questions in this study guide pertain to course lecture videos.

Syllabus Question 1.2

Set 0 A

Based on the syllabus, in what type of class would Dr. Sanford expect to receive the most complaints from students about not receiving enough feedback?

- A. If the students are not given any study guides or any direction on how to study for tests.
- B. If the students are given access to a pool of questions used to create the tests and given many resources to test their own knowledge.
- C. If student grades are determined by a very small number of tests so that they rarely receive feedback (for example, if grades are determined by just two midterm tests and a final exam).
- D. If the students are not given any tools they can use to develop their own judgment of learning.

Syllabus Question 1.3

Set 0 A

According to the syllabus, which of the following describes what is most likely to happen if it is especially easy for students to obtain correct answers?

- A. Students may become more motivated to develop their own judgment of learning.
- B. Students may become overly focused on understanding concepts in situations where they need to use more rote memorization.
- C. Students may increase their use of lecture videos to address identified areas of weakness.
- D. Students may tend to overestimate how well they understand the material.

Syllabus Question 1.4

Set 0 A

According to the syllabus, students should obtain item-by-item feedback through all the following sources EXCEPT for which of the following.

- A. pre-learning-assessment practice quizzes
- B. learning assessments
- C. homework assignments
- D. projects
- E. asking questions in class

Syllabus Question 1.5

Set 0 A

According to the syllabus, which of the following is true about the reliability of learning assessments?

- A. The learning assessments you take in this class are likely to be less reliable than tests you take in other classes.
- B. It would be normal if random factors cause a student's score to be 8 points higher or 8 points lower on a single learning assessment.
- C. A score from a single learning assessment may sometimes be a more reliable estimate of your performance than averaging across multiple scores.
- D. You should contact Dr. Sanford or a Teaching Assistant if you have reason to believe that random factors caused you to miss 2 points or more on a learning assessment.

Syllabus Question 1.6

Set 0 A

Imagine a student became sick and needed to miss a learning assessment. Which of the following is the first thing this student should do?

- A. Send a notification email to document@PsyBU.com that contains a copy of a doctors note.
- B. Before 5:00 on the day of the absence, complete the “Request for makeup” form and submit it to document@PsyBU.com.
- C. Before the start of class, send a notification email to document@PsyBU.com stating the reason for the absence.
- D. Contact Dr. Sanford or a Teaching Assistant and ask if a makeup assessment can be arranged.

Syllabus Question 1.7

Set 0 A

Which of the following is true about the “Request for Makeup” form?

- A. Submitting the form requires students to attach documenting evidence.
- B. If a student misses several learning assessments in a row, the student only needs to submit one form for all the absences.
- C. If a student directly contacts Dr. Sanford or a Teaching Assistant to explain an absence, it is not necessary to complete the “Request for Makeup” form.
- D. Completing the form requires students to provide answers to study guide questions.

Syllabus Question 1.8

Set 0 A

Which of the following would incur the largest penalty on a makeup learning assessment?

- A. Submitting a scheduling request more than one week late
- B. Requesting more than 4 makeup learning assessments
- C. Requesting a makeup session that is NOT the earliest available session
- D. Making minor, careless errors on the “Request for Makeup” form

Syllabus Question 1.9

Set 0 A

Which of the following is true about submitting a “waiver request?”

- A. If the waiver involves submission of a late assignment, the student should wait until the waiver is approved before submitting the assignment.
- B. Students will typically receive an answer to waiver requests within one week of submission.
- C. Students may submit waiver requests to take a makeup learning assessment outside of a regularly scheduled makeup session.
- D. If the waiver involves taking more than 4 makeup assessments, it needs to include documenting evidence.
- E. Students may submit waiver requests to take a makeup learning assessment at home.

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Question 2

Set 1 A

Professor Cid observed that, “the patient was in good condition, talked relevantly and coherently, and was correctly oriented.” This is an example of _____.

Check **ALL** the answers that are true.

- A. A quantified observation
- B. An observation that has NOT been quantified
- C. A variable
- D. A categorical variable
- E. A numerically scaled variable

Question 3

Set 1 A

A sample of patients were given scores on the Brief Psychiatric Rating Scale. This is an example of _____ .

Check **ALL** the answers that are true.

- A. Quantified observations
- B. Observations that have NOT been quantified
- C. A variable
- D. A categorical variable
- E. A numerically scaled variable

Question 4

Set 1 A

In a sample of patients, some were in a treatment group and others were in a control group. This is an example of _____ .

Check **ALL** the answers that are true.

- A. A quantified observation
- B. Observations that have NOT been quantified
- C. A variable
- D. A categorical variable
- E. A numerically scaled variable

Question 5

Set 1 A

Correctly spell the term that has to do with the extent to which the same amount of something always produces the same assessed numerical value regardless of when or how it is assessed.

Correctly spell the term that has to do with the extent to which a scale measures what it is supposed to measure.

Question 6

Set 1 A

Correctly spell the term that goes in each blank.

A study where scores for all the variables are determined by assessment and there are NO manipulated variables is called _____ research.

A study that has at least one variable with scores determined by an assessment and at least one manipulated variable produced by the researcher forcing cases into groups is called _____ research.

Question 7

Set 1 A

Fifty people received a treatment for anxiety. Their level of anxiety was assessed both before the treatment and after the treatment. This is an example of _____ .

Check **ALL** the answers that are true.

- A. A study using random assignment
- B. A type of research that was specifically described in class as being limited in its ability to determine cause.
- C. A type of research that was specifically described in class as being limited in its ability to explain natural phenomena.
- D. A study that includes a manipulated variable
- E. Correlational research
- F. A clinical trial

Question 8

Set 1 A

A researcher conducts a clinical trial. This is an example of _____ .

Check **ALL** the answers that are true.

- A. A study using random assignment
- B. A type of research that was specifically described in class as being limited in its ability to determine cause.
- C. A type of research that was specifically described in class as being limited in its ability to explain natural phenomena.
- D. A study that includes a manipulated variable
- E. Correlational research
- F. A study that includes a categorical variable

Question 9

Set 1 A

Correctly spell the term for a numerical value based on data from all the cases in a **population** that describes something about the **population**.

Question 10

Set 1 A

Correctly spell the term that goes in each blank.

To address the question, “how big is the effect?” we need to calculate an effect size. This will be a(an) _____ statistic.

To address the question, “could the results be random chance?” we need to calculate a(an) _____ statistic.

Question 11**Set 1 B**

$$x_i = \bar{x} + e_i$$

What is e_i ?

- A. The score you expect to get for case i .
- B. The actual score for case i .
- C. How much case i deviates from the mean.
- D. The mean of all the “ e ” values in the sample.

Question 12**Set 1 B**

$$x_i = \bar{x} + e_i$$

What is $\bar{x}$?

- A. The score you expect to get for case i .
- B. The actual score for case i .
- C. How much case i deviates from the mean.
- D. The mean of all the “ e ” values in the sample.

Question 13**Set 1 B**

$$x_i = \bar{x} + e_i$$

Write an alternate version for this linear equation that includes a term for the **predicted** score on x.

Question 14**Set 1 B**

$$\hat{x}_i = \bar{x}$$

Write an alternate version for this linear equation that does NOT use the “hat” symbol.

Question 15

Set 1 B

Write the definition of a mean.

Note: Your definition should state the most important things to know about a mean.

You may write the lyrics to the song or paraphrase the key point that was made in the song.

Question 16

Set 1 B

Write the definition of a standard deviation.

Note: Your definition should state the most important things to know about a standard deviation.

You may write the lyrics to the song or paraphrase the key points that were made in the song.

Question 17

Set 1 B

Write the formula for the sum of squares of variable x .
(Included any necessary subscripts in this formula.)

Use the letters “ss” to indicate “sum of squares” and write the formula for a variance of variable x .

Use the letters “ss” to indicate “sum of squares” and write the formula for a standard deviation of variable x .

Question 18

Set 1 B

In the process of computing a standard deviation, Zelda lost one degree of freedom. What did Zelda do that **caused** her to lose one degree of freedom?

- A. She omitted one person from the sample.
- B. She used a statistic that was smaller than its expected value.
- C. She used a biased formula to calculate the standard deviation.
- D. She divided the variance by $n-1$.
- E. She used squared deviations rather than absolute deviations.
- F. She calculated the mean

Question 19

Set 1 B

Correctly spell the **statistical term** that is a **single word** and that goes in the blank. Bruno computed a standard deviation, but he accidentally used “n” rather than the degrees of freedom in his formula. This means that the value he obtains will be _____.

Question 20

Set 1 B

Correctly spell the term used when the expected value is not equal to the parameter.

Question 21

Set 1 B

Correctly spell the term that indicates the average of all the values for a population of values from which a single value can be randomly sampled.

Question 22

Set 1 C

Write the formula for a z -score.

Use the letter “ z ” to indicate a z -score and write the formula for a T -score.

Write the formula for a centered score.

Question 23

Set 1 C

Correctly spell the term for the thing that has a mean of 0 and a standard deviation of 1.

Correctly spell the term for the thing that has a mean of 50 and a standard deviation of 10.

Question 24

Set 1 C

A z-score is equal to a _____ divided by a _____.

What word goes in the first blank?

What word goes in the second blank?

- A. mean
- B. centered score
- C. difference score
- D. expected score
- E. standard deviation

Question 25

Set 1 C

If scores in a sample are normally distributed, what is the probability of randomly selecting someone from the sample that has a z -score of **2 or lower**?

If scores in a sample are normally distributed, what is the probability of randomly selecting someone from the sample that has a z -score of **1 or higher**?

Question 26

Set 2 A

Write the definition of a d -statistic.

Note: Your definition should state the most important things to know about a d -statistic.

You may write the lyrics to the song or paraphrase the key points that were made in the song.

Question 27

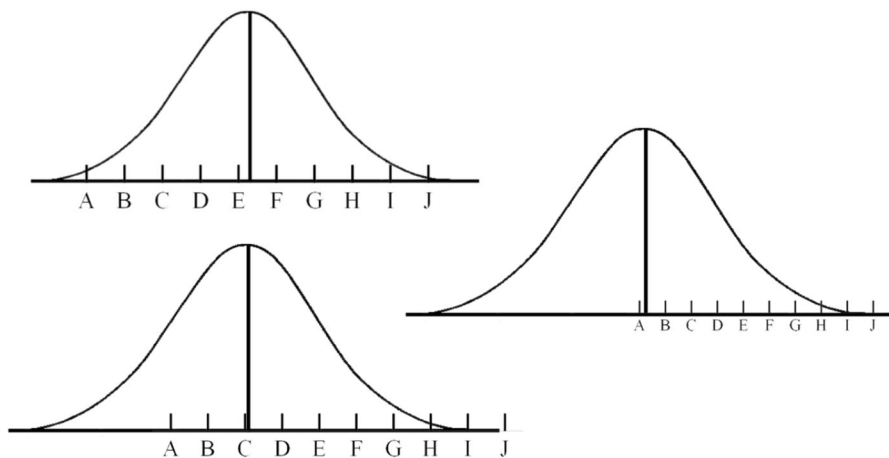
Set 2 A

Write the formula for a d -statistic.

Question 28

Set 2 A

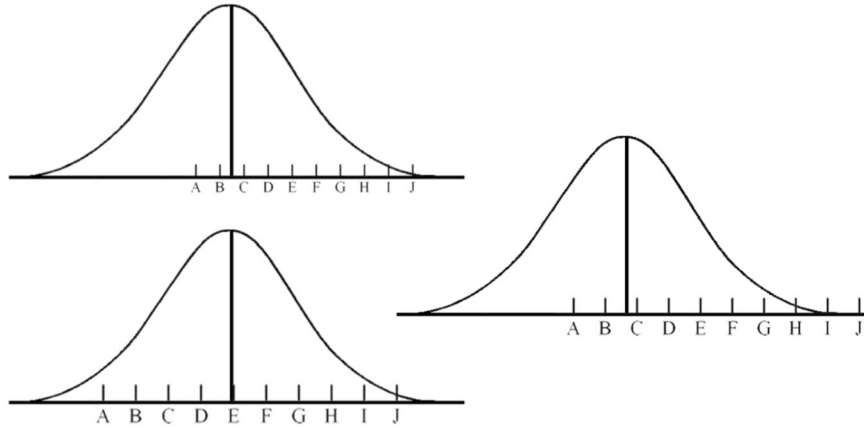
In each figure, which letter is 1 standard deviation above the mean?



Question 29

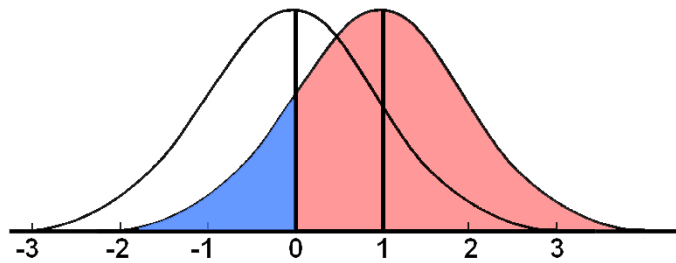
Set 2 A

In each figure, which letter is 2 standard deviations above the mean?



Question 30

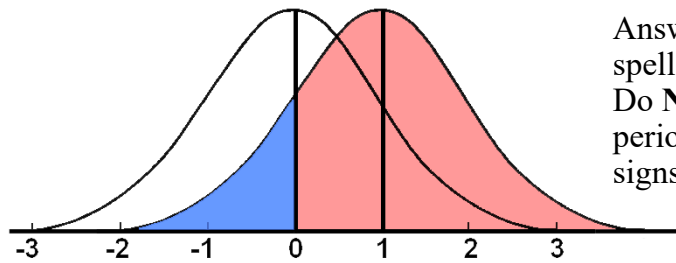
Set 2 A



What is the value of the d -statistic depicted in this figure?

Question 31

Set 2 A



Answers must be spelled correctly. Do **NOT** type any periods, percentage signs, or spaces.

Type a single word to fill in the blank.

This is a _____ effect.

In each blank, type a 2-digit number (and nothing else).

With **no** treatment your probability

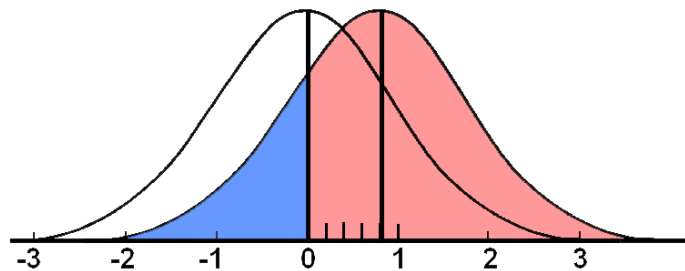
of superiority is _____ percent,

with treatment it increases to _____ percent,

which is an increase of _____ percent.

Question 32

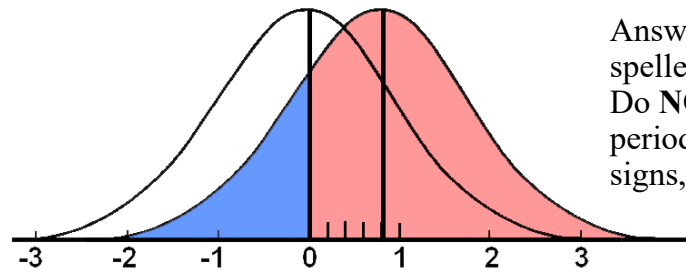
Set 2 A



What is the value of the d -statistic depicted in this figure?

Question 33

Set 2 A



Answers must be spelled correctly. Do **NOT** type any periods, percentage signs, or spaces.

Type a single word to fill in the blank.

This is a _____ effect.

In each blank, type a 2-digit number (and nothing else).

With **no** treatment your probability

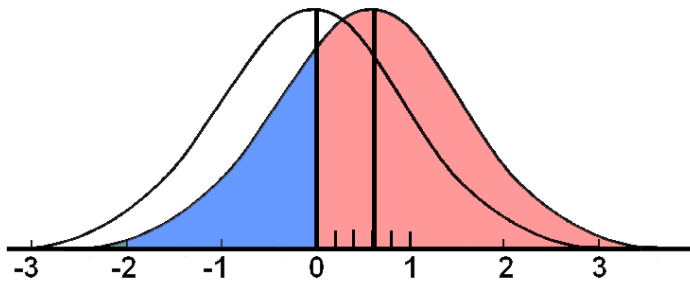
of superiority is _____ percent,

with treatment it increases to _____ percent,

which is an increase of _____ percent.

Question 34

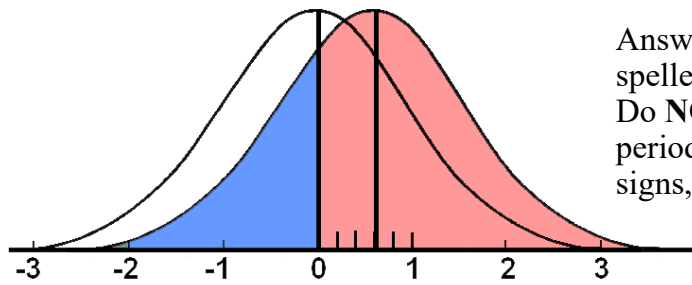
Set 2 A



What is the value of the d -statistic depicted in this figure?

Question 35

Set 2 A



Answers must be spelled correctly. Do **NOT** type any periods, percentage signs, or spaces.

Type a single word to fill in the blank.

This is a _____ effect.

In each blank, type a 2-digit number (and nothing else).

With **no** treatment your probability

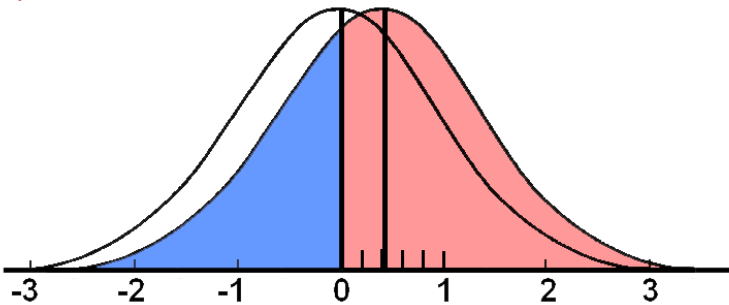
of superiority is _____ percent,

with treatment it increases to _____ percent,

which is an increase of _____ percent.

Question 36

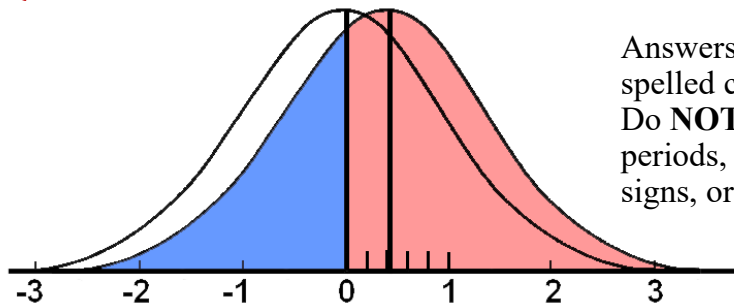
Set 2 A



What is the value of the d -statistic depicted in this figure?

Question 37

Set 2 A



Answers must be spelled correctly.
Do **NOT** type any periods, percentage signs, or spaces.

Type a single word to fill in the blank.

This is a _____ effect.

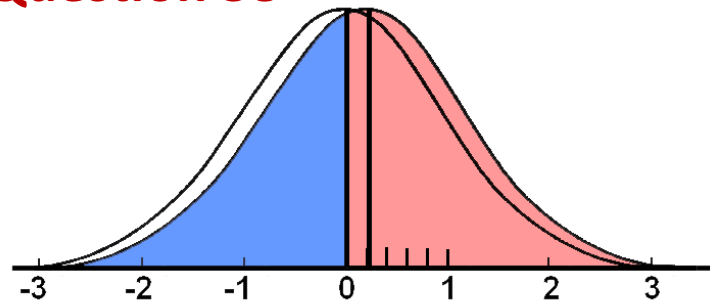
In each blank, type a 2-digit number (and nothing else).

With **no** treatment your probability of superiority is _____ percent,

with treatment it increases to _____ percent, which is an increase of _____ percent.

Question 38

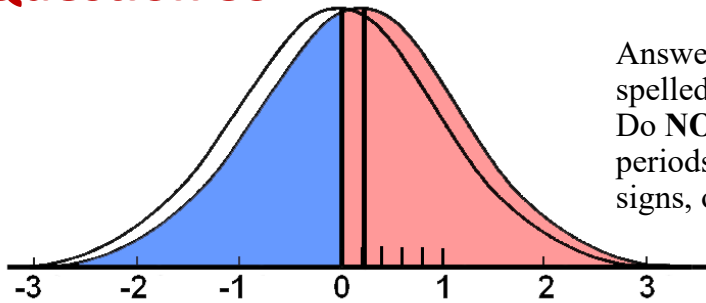
Set 2 A



What is the value of the d -statistic depicted in this figure?

Question 39

Set 2 A



Answers must be spelled correctly. Do **NOT** type any periods, percentage signs, or spaces.

Type a single word to fill in the blank.

This is a _____ effect.

In each blank, type a 1- or 2-digit number (and nothing else).

With **no** treatment your probability

of superiority is _____ percent,

with treatment it increases to _____ percent,

which is an increase of _____ percent.

Question 40

Set 2 B

Write the formula for a variance **and** the formula for a covariance.

You may write these formulas without using subscripts.

Question 41

Set 2 B

Think about the formulas for variance and covariance. Which of the following is **NOT** part of the formula for a **covariance**?

- A. squaring values
- B. multiplying x values times y values
- C. centering scores on x
- D. dividing by $n-1$
- E. summing values

Question 42

Set 2 B

A correlation is equal to _____ divided by _____ .

What goes in the first blank?

What goes in the second blank?

- A. A difference between means
- B. A centered score
- C. A covariance
- D. The highest possible value a covariance could take
- E. The highest possible value a standard deviation could take
- F. The cross product of two centered scores

Question 43

Set 2 B

Use “cov_xy” as the symbol for the covariance between x and y and write the formula for a correlation.

Question 44

Set 2 C

Write a linear equation that includes an “ e ” where the average deviation of $|e|$ is the **standard deviation**.

(Note: The equation should include subscripts.)

Question 45

Set 2 C

Fill in each blank by typing correctly spelled words that match the words used in the googly-eyed sheep table (the one titled, “Types of distributions”).

A **standard deviation** pertains to:

- Observations of different _____ from a single _____.
- Deviations around a _____.

Question 46

Set 2 C

[equation 1] $y_{is} = (\text{prediction}_s \text{ based on } x_{is}) + e_{is}$

[equation 2] $x_{is} = \bar{x}_s + e_{is}$

The standard deviation _____

The standard error of estimate _____

- A. is e_{is} from equation 1.
- B. is the averaged (absolute) size of e_{is} from equation 1.
- C. is e_{is} from equation 2.
- D. is the averaged (absolute) size of e_{is} from equation 2.

Question 47

Set 2 C

Use the letter “w” to indicate the term:
(prediction_s based on x_{is})

Write a linear equation that includes an “e” where the average deviation of $|e|$ is the **standard error of estimate**.

Note: although you should *not* add subscripts to the letter “w,” other parts of the equation **DO** need subscripts.

Question 48

Set 2 C

Correctly spell the terms that need to be placed in the blanks to produce a linear equation where the average deviation of $|e|$ is the **standard error of estimate**. In typing your answer, use an underscore to indicate subscripts as you do when typing formulas.

y_is = (_____ based on _____) + e_is

Question 49

Set 2 C

Fill in each blank by typing correctly spelled words that match the words used in the googly-eyed sheep table (the one titled, “Types of distributions”).

A **standard error of estimate** pertains to:

- Observations of different _____ from a single _____ .
- Deviations around a _____ .

Question 50

Set 2 C

Write the definition of a standard error of estimate.

Your definition should state the most important things to know about a standard error of estimate.

You may write the lyrics to the song or paraphrase the key points that were made in the song.

Question 51

Set 2 D

What is a linear equation that includes an “ e ” where the average deviation of $|e|$ is the **standard error of measure**?

(Note: The equation should include subscripts)

Question 52

Set 2 D

Fill in each blank by typing correctly spelled words that match the words used in the googly-eyed sheep table (the one titled, “Types of distributions”).

A **standard error of measure** pertains to:

- Observations of different _____ from a single _____ .
- Deviations around a _____ .

Question 53

Set 2 D

Write the definition of the standard error of measure.

Your definition should state the most important things to know about a standard error of measure.

You may write the lyrics to the song or paraphrase the key points that were made in the song.

Question 54

Set 2 D

In the equation, $x = \text{true} + e$, the “ e ” needs a subscript to indicate it belongs to:

- A. sample “s”
- B. individual “i” from sample “s”
- C. assessment occasion “a” for individual “i”
- D. assessment occasion “a” from sample “s”

If Cronbach’s α is .6, it indicates that

- A. people typically deviate from their true scores by 60%.
- B. the standard error of measure is .6
- C. 60% of the variance is measurement error
- D. 60% of the variance is true variance

Question 55

Set 2 E

Write the linear equation that includes an “ e ” where the average deviation of $|e|$ is the **standard error of a mean**.

Note: The equation should include subscripts.

If you need to use Greek letters, remember that:

μ = mu

β = beta

ρ = rho

Question 56

Set 2 E

Fill in each blank by typing correctly spelled words that match the words used in the googly-eyed sheep table (the one titled, “Types of distributions”).

A **standard error of a mean** pertains to:

- Observations of different _____ from a single _____ .
- Deviations around a _____ .

Question 57

Set 2 E

Write the linear equation that includes an “ e ” where the average deviation of $|e|$ is the **standard error of a difference between means**?

Note: The equation should include subscripts.

If you need to use Greek letters, remember that:

μ = mu

β = beta

ρ = rho

Question 58

Set 2 E

Fill in each blank by typing correctly spelled words that match the words used in the googly-eyed sheep table (the one titled, “Types of distributions”).

A **standard error of a difference between means** pertains to:

- Observations of different _____ from a single _____ .
- Deviations around a _____ .

Question 59

Set 2 E

Write the linear equation that includes an “ e ” where the average deviation of $|e|$ is the **standard error of a beta**.

Note: The equation should include subscripts.

The symbol for a beta is “ β .”

If you need to use Greek letters, remember that:

μ = mu

β = beta

ρ = rho

Question 60

Set 2 E

Fill in each blank by typing correctly spelled words that match the words used in the googly-eyed sheep table (the one titled, “Types of distributions”).

A **standard error of a beta** pertains to:

- Observations of different _____ from a single _____ .
- Deviations around a _____ .

Question 61

Set 2 E

Write the linear equation that includes an “ e ” where the average deviation of $|e|$ is the **standard error of a correlation**?

Note: The equation should include subscripts.

The symbol for a correlation is “ r .”

If you need to use Greek letters, remember that:

μ = mu

β = beta

ρ = rho

Question 62

Set 2 E

Fill in each blank by typing correctly spelled words that match the words used in the googly-eyed sheep table (the one titled, “Types of distributions”).

A **standard error of a correlation** pertains to:

- Observations of different _____ from a single _____ .
- Deviations around a _____ .

Question 63

Set 2 E

Correctly spell a single word that goes in each blank.

When writing linear equations (for the standard deviation, standard error of estimate, standard error of measure, and standard error of a statistic), you need to use the subscripts, “i,” “s,” “a,” and “p.”

“i” stands for _____ .

“s” stands for _____ .

“a” stands for _____ .

“p” stands for _____ .

Question 64

Set 2 E

Write the definition of a standard error of a statistic.

Your definition should state the most important things to know about a standard error of a statistic.

You may write the lyrics to the song or paraphrase the key points that were made in the song.

Question 65

Set 2 E

The **smaller** the sample. . .

- A. The **more** the standard error will have a normal distribution.
- B. The **smaller** the size of e_{sp} .
- C. The **smaller** the standard error of a statistic.
- D. The **larger** the standard error of a statistic.

The standard error of a correlation indicates how far correlations from different studies typically deviate from ____ .

- A. the population parameter
- B. the sample mean
- C. the predicted values
- D. the true scores.

Question 66

Set 2 E

Correctly spell the term that goes in the blank.

The size of the standard error of a statistic is primarily influenced by the size of the _____ .

Question 67

Set 2 E

Correctly spell the term that goes in the blank.

Standard error of a statistic will be almost perfectly _____ in shape.

Question 68

Set 2 F

Write the definition of a t -value.

Your definition should state the most important things to know about a t -value.

You may write the lyrics to the song or paraphrase the key points that were made in the song.

Question 69**Set 2 F**

Using the words “parameter” and “statistic,” write the formula for a t -value.

Question 70**Set 2 F**

Write the definition of “significance.”

Your definition should state the most important things to know about significance.

You may write the lyrics to the song or paraphrase the key points that were made in the song.

Question 71

Set 2 F

“If we knew the population parameter for this, the distribution would be a z and not a t .” To which part of the t -value formula does this statement apply?

- A. statistic – parameter
- B. statistic
- C. parameter
- D. $sd_{\text{statistic}}$

If the p value is .08, it means that

- A. there is a 8% probability that the null hypothesis is true.
- B. we can be 92% confident that the research hypothesis is true.
- C. if the null hypothesis were true, we would get an effect this large only 8% of the time.
- D. the results are significant.

Question 72

Set 3 A

For a single sample estimate of a **single mean**,

Write the linear equation for **variance in a sample**.

Note: The equation should include subscripts.

Question 73

Set 3 A

For a single sample estimate of a **single mean**,

Write the linear equation for **sampling error variance in a population**.

Note: The equation should include subscripts.

If you need to use Greek letters, remember that:

μ = mu

β = beta

ρ = rho

Question 74

Set 3 A

Correctly spell the terms that go in the blanks.

The equation $x_{is} = \bar{x}_s + e_{is}$ includes two types of components.

First, a _____ is a value that can vary between people in sample.

Second, a _____ is a value that can vary between studies in a population.

Question 75

Set 3 A

Consider the two equations below:

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

Which of the following are **variables**? Select ALL the correct answers.

A. x_{is}

B. μ_p

C. $\bar{x}_s$

D. e_{sp}

E. e_{is}

Question 76

Set 3 A

Consider the two equations below:

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

Which of the following are **parameter estimates**? Select ALL the correct answers.

A. x_{is}

B. μ_p

C. $\bar{x}_s$

D. e_{sp}

E. e_{is}

Question 77

Set 3 A

Consider the two equations below:

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

Which of the following answers the **descriptive question** and would provide the primary result we want to know when conducting a study?

A. x_{is}

B. μ_p

C. $\bar{x}_s$

D. e_{sp}

E. e_{is}

Question 78

Set 3 A

Consider the two equations below:

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

Which of the following is the primary component that we need to use to answer the **inferential question** ?

A. x_{is}

B. μ_p

C. $\bar{x}_s$

D. e_{sp}

E. e_{is}

Question 79

Set 3 A

Consider the two equations below:

$$x_{is} = \bar{x}_s + e_{is} \qquad \bar{x}_{sp} = \mu_p + e_{sp}$$

- A. The average deviation of $|e_{is}|$.
- B. The average deviation of $|e_{sp}|$.

Match each
definition above
with the appropriate
term in the box.

Possible terms:

standard error of measure
standard deviation
standard error of estimate
standard error of a statistic
z-score
t-value

Question 80

Set 3 A

Consider the equation: $\bar{x}_{sp} = \mu_p + e_{sp}$.

Assume you want to know the standardized e_{sp} for a particular study assuming the null hypothesis is true.

This can be answered with:

- A. a descriptive statistic
- B. the standard error of the mean
- C. a t -value
- D. the standard error of estimate

Question 81

Set 3 A

The standard of the error of the mean will be the largest when the **standard deviation** is _____ and the **sample size** is _____ .

- A. large, large
- B. large, small
- C. small, large
- D. small, small

Which of the following will produce the **largest** standard error of the mean?

- A. sd = 2, and the sample size is small
- B. sd = 2, and the sample size is big
- C. sd = 8, and the sample size is small
- D. sd = 8, and the sample size is big

Question 82

Set 3 A

Linear equations

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

$$\bar{x}_s = 10 \quad sd = 2.12$$

$$sd_{\bar{x}} = \frac{2.12}{\sqrt{5}} = .95$$

$$n = 5 \quad t = \frac{10}{.95} = 10.53$$

If H0 is true, how many standard error units did this study deviate from the null population parameter?

What is the average deviation of $|e_{is}|$?

Question 83

Set 3 A

Linear equations

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

$$\bar{x}_s = 10 \quad sd = 2.12$$

$$sd_{\bar{x}} = \frac{2.12}{\sqrt{5}} = .95$$

$$n = 5 \quad t = \frac{10}{.95} = 10.53$$

What is the average deviation of $|e_{sp}|$?

If H_0 is true, what is the standardized e_{sp} for this particular study?

Question 84

Set 3 A

Linear equations

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

$$\bar{x}_s = 10 \quad sd = 2.12$$

$$sd_{\bar{x}} = \frac{2.12}{\sqrt{5}} = .95$$

$$n = 5 \quad t = \frac{10}{.95} = 10.53$$

Which numerical value answers the **descriptive question** and provides the primary result the researcher wanted to know when conducting this study?

Question 85

Set 3 A

Linear equations

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

$$\bar{x}_s = 10 \quad sd = 2.12$$

$$sd_{\bar{x}} = \frac{2.12}{\sqrt{5}} = .95$$

$$n = 5 \quad t = \frac{10}{.95} = 10.53$$

To look up the probability of these results assuming the null hypothesis is true, you need to use two numerical values. One of these values is equal to 4. What is the other numerical value you need to use?

Question 86

Set 3 B

	x_1	x_2	$x - x$
	9	8	1
	8	5	3
	7	4	3
	4	3	1
	3	1	2

Results from a paired samples study

$$\bar{x}_{\text{diff}} = 2 \quad sd_{\text{diff}} = 1$$

Use numerical values (and NOT letters or subscripts) and write the linear equations for:

- variance in a sample using values for Case 2
- sampling error variance in a population assuming H_0 is true

Question 87

Set 3 B

	x_1	x_2
	9	7
	8	4
	6	4
	5	1

$$\bar{x}_1 = 7$$

$$\bar{x}_2 = 7$$

$$\bar{x}_{diff} = 3$$

Assume you do a paired samples study and get this dataset with 4 people.

To calculate sd_{diff} , you would calculate the standard deviation using which set scores?

- A. 9, 8, 6, 5
- B. 7, 4, 4, 1
- C. 9, 8, 6, 5, 7, 4, 4, 1
- D. 2, 4, 2, 4
- E. 7, 7, 3

Question 88

Set 3 B

	x_1	x_2
	9	5
	8	5
	6	3
	5	3

Assume you do a paired samples study and get this dataset with 4 people.

What is $\bar{x}_{diff}$?

Question 89

Set 3 B

Paired samples study results

Linear equations

$$x_{\text{diff } is} = \bar{x}_{\text{diff } s} + e_{\text{diff } is}$$

$$\bar{x}_{\text{diff } sp} = \mu_{\text{diff } p} + e_{\text{diff } sp}$$

$$\bar{x}_{\text{diff}} = 2 \quad sd_{\text{diff}} = 1$$

$$sd_{\bar{x} \text{ difference}} = \frac{1}{\sqrt{5}} = .447$$

$$n = 5 \quad t = \frac{2}{.447} = 4.47$$

If H_0 is true, how many standard error units did this study deviate from the null population parameter?

What is the average deviation of $|e_{\text{diff } is}|$?

Question 90

Set 3 B

Paired samples study results

Linear equations

$$x_{\text{diff } is} = \bar{x}_{\text{diff } s} + e_{\text{diff } is}$$

$$\bar{x}_{\text{diff } sp} = \mu_{\text{diff } p} + e_{\text{diff } sp}$$

$$\bar{x}_{\text{diff}} = 2 \quad sd_{\text{diff}} = 1$$

$$sd_{\bar{x} \text{ difference}} = \frac{1}{\sqrt{5}} = .447$$

$$n = 5 \quad t = \frac{2}{.447} = 4.47$$

What is the average deviation of $|e_{\text{diff } sp}|$?

If H_0 is true, what is the standardized $e_{\text{diff } sp}$ for this particular study?

Question 91

Set 3 B

Paired samples study results

Linear equations

$$x_{\text{diff } is} = \bar{x}_{\text{diff } s} + e_{\text{diff } is}$$

$$\bar{x}_{\text{diff } sp} = \mu_{\text{diff } p} + e_{\text{diff } sp}$$

$$\bar{x}_{\text{diff}} = 2 \quad sd_{\text{diff}} = 1$$

$$sd_{\bar{x} \text{ difference}} = \frac{1}{\sqrt{5}} = .447$$

$$n = 5 \quad t = \frac{2}{.447} = 4.47$$

Which numerical value answers the **descriptive question** and provides the primary result the researcher wanted to know when conducting this study?

Question 92

Set 3 B

Paired samples study results

Linear equations

$$x_{\text{diff } is} = \bar{x}_{\text{diff } s} + e_{\text{diff } is}$$

$$\bar{x}_{\text{diff } sp} = \mu_{\text{diff } p} + e_{\text{diff } sp}$$

$$\bar{x}_{\text{diff}} = 2 \quad sd_{\text{diff}} = 1$$

$$sd_{\bar{x} \text{ difference}} = \frac{1}{\sqrt{5}} = .447$$

$$n = 5 \quad t = \frac{2}{.447} = 4.47$$

To look up the probability of these results assuming the null hypothesis is true, you need to use two numerical values. One of these values is equal to 4. What is the other numerical value you need to use?

Question 93

Set 3 B

Which of the following is true about a paired sample correlation?

- A. It indicates how much the rank order of cases were stable across different assessment points.
- B. As the mean difference score gets larger, the paired sample correlation gets **larger**.
- C. As the mean difference score gets larger, the paired sample correlation gets **smaller**.
- D. An example of a paired sample correlation could be the correlation between self-compassion and psychological distress.
- E. The paired sample correlation will always equal one.

Question 94

Set 3 C

For a two-samples study investigating a **difference between two means**, write the linear equation for **variance in a sample**.

Note: The equation should include subscripts.

Question 95

Set 3 C

For a two-samples study investigating a **difference between two means**, there are two equations for sampling error variance in a population. To write the **most important** of these two equations, you would need to include the term “e_{1-0sp}.”

Write this linear equation for sampling error variance in a population that includes the term “e_{1-0sp}.”

Note: The equation should include subscripts.

If you need to use Greek letters, remember that:

$$\mu = \text{mu}$$

$$\beta = \text{beta}$$

$$\rho = \text{rho}$$

Question 96

Set 3 C

For a two-samples study investigating a **difference between two means**, there are two equations for sampling error variance in a population. To write the **least** important of these two equations, you would need to include the term “e_{0sp}.”

Write this linear equation for sampling error variance in a population that includes the term “e_{0sp}.”

Note: The equation should include subscripts.

If you need to use Greek letters, remember that:

$$\mu = \text{mu}$$

$$\beta = \text{beta}$$

$$\rho = \text{rho}$$

Question 97

Set 3 C

Person	y	x
1	8	1
2	6	1
3	7	1
4	7	1
5	3	0
6	2	0
7	4	0
8	3	0

Results from a study investigating the difference between two means

$$\bar{y}_1 = 7$$

$$\bar{y}_0 = 3$$

$$\bar{y}_1 - \bar{y}_0 = 4$$

Use numerical values (and NOT letters or subscripts) and write the linear equation for **variance in a sample** using values for Person 1.

Question 98

Set 3 C

Person	y	x
1	8	1
2	6	1
3	7	1
4	7	1
5	3	0
6	2	0
7	4	0
8	3	0

$$\bar{y}_1 = 7 \quad \bar{y}_0 = 3 \quad \bar{y}_1 - \bar{y}_0 = 4$$

Given this example, there are two equations for sampling error variance in a population, and one is more important than the other.

Use numerical values (and NOT letters or subscripts) and write the **most** important linear equation for **sampling error variance in a population** assuming H_0 is true.

Question 99

Set 3 C

Person	y	x
1	8	1
2	6	1
3	7	1
4	7	1
5	3	0
6	2	0
7	4	0
8	3	0

$$\bar{y}_1 = 7 \quad \bar{y}_0 = 3 \quad \bar{y}_1 - \bar{y}_0 = 4$$

Given this example, there are two equations for sampling error variance in a population, and one is more important than the other.

Use numerical values (and NOT letters or subscripts) and write the **least** important linear equation for **sampling error variance in a population** assuming H_0 is true.

Question 100

Set 3 C

Consider the three equations below:

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-sp}$$

Which of the following are variables? Select ALL the correct answers.

$$y_{is} \quad \bar{y}_{0s} \quad (\bar{y}_{1s} - \bar{y}_{0s})$$

$$x_{is} \quad e_{is} \quad \mu_{0p}$$

$$(\mu_{1p} - \mu_{0p}) \quad e_{1-sp}$$

Question 101

Set 3 C

Consider the three equations below:

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

Which of the following are parameter estimates?

Select ALL the correct answers.

$$y_{is} \quad \bar{y}_{0s} \quad (\bar{y}_{1s} - \bar{y}_{0s})$$

$$x_{is} \quad e_{is} \quad \mu_{0p}$$

$$(\mu_{1p} - \mu_{0p}) \quad e_{1-0sp}$$

Question 102

Set 3 C

Consider the three equations below:

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

Which of the following answers the **descriptive question** and would provide the primary result we want to know when conducting a study?

$$y_{is} \quad \bar{y}_{0s} \quad (\bar{y}_{1s} - \bar{y}_{0s})$$

$$x_{is} \quad e_{is} \quad \mu_{0p}$$

$$(\mu_{1p} - \mu_{0p}) \quad e_{1-0sp}$$

Question 103

Set 3 C

Two-samples study results

Linear equations

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

$$n = 8$$

$$\bar{y}_1 = 3 \quad \bar{y}_0 = 8$$

$$sd_1 \approx .82 \quad sd_0 \approx .82$$

$$\bar{y}_1 - \bar{y}_0 = -5$$

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{.667}{4} + \frac{.667}{4}} \approx .58$$

$$t = \frac{-5}{.58} \approx -8.62$$

If H0 is true, how many standard error units did this study deviate from the null population parameter?

Question 104

Set 3 C

Two-samples study results

Linear equations

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

$$n = 8$$

$$\bar{y}_1 = 3 \quad \bar{y}_0 = 8$$

$$sd_1 \approx .82 \quad sd_0 \approx .82$$

$$\bar{y}_1 - \bar{y}_0 = -5$$

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{.667}{4} + \frac{.667}{4}} \approx .58$$

$$t = \frac{-5}{.58} \approx -8.62$$

What is the average deviation of $|e_{1-0sp}|$?

If H0 is true, what is the standardized e_{1-0sp} for this particular study?

Question 105

Set 3 C

Two-samples study results

Linear equations

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

$$n = 8$$

$$\bar{y}_1 = 3 \quad \bar{y}_0 = 8$$

$$sd_1 \approx .82 \quad sd_0 \approx .82$$

$$\bar{y}_1 - \bar{y}_0 = -5$$

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{.667}{4} + \frac{.667}{4}} \approx .58$$

$$t = \frac{-5}{.58} \approx -8.62$$

Which numerical value answers the **descriptive question** and provides the primary result the researcher wanted to know when conducting this study?

Question 106

Set 3 C

Two-samples study results

Linear equations

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

$$n = 8$$

$$\bar{y}_1 = 3 \quad \bar{y}_0 = 8$$

$$sd_1 \approx .82 \quad sd_0 \approx .82$$

$$\bar{y}_1 - \bar{y}_0 = -5$$

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{.667}{4} + \frac{.667}{4}} \approx .58$$

$$t = \frac{-5}{.58} \approx -8.62$$

To look up the probability of these results assuming the null hypothesis is true, you need to use two numerical values. One of these values is equal to 6. What is the other numerical value you need to use?

Question 107

Set 3 D

If you do a study investigating the difference in means between two groups, you have a few options for reporting the effects size. Which option can be interpreted most easily (and is the default option for use in this class)?

A. p value

B. $\frac{\bar{y}_1 - \bar{y}_0}{sd_{pooled}}$

C. $\frac{\text{statistic}}{sd_{\text{statistic}}}$

D. $\sqrt{\frac{t^2}{t^2 + df}}$

Question 108

Set 3 D

When using a t -value to test for a difference between means, two assumptions need to be true for the test to be valid. One pertains to heterogeneity of variance. Which of the following is true of this assumption?

- A. This assumption is more important than the assumption regarding independence of observations.
- B. A study needs to have heterogeneity of variance to produce a valid t -test result.
- C. This assumption would be violated if the same person participated in a study multiple times.
- D. A significant result from a Levene's test indicates that this assumption has been violated.

Question 109

Set 3 D

If the critical value is 1.98, this means that _____ percent of values in a t -distribution will be more extreme (in either a positive or negative direction) than 1.98.

Question 110

Set 3 D

The mean for group one is 75 and the mean for group zero is 50 (therefore the difference between means is 25). The standard error of difference is 5 and the critical value (for $p = .05$) is 2.

What is the lower bound of the 95% confidence interval for the difference between means?

Question 111

Set 3 D

Fill in the blanks with correctly spelled terms.

Class lecture included a description of how you can provide “five pieces of information” to answer research questions. First, you provide the name and the value of a(an) _____ statistic, and this indicates the size of the effect. Then, you provide the name and the value of a(an) _____ statistic along with a _____ value.

Question 112

Set 4 A

If a correlation is .5, what is the percent of variance explained?

Question 113

Set 4 A

Consider the following example:

Zara's score on $x = 5$

Zara's score on $y = 23$

$$b_0 = 10$$

$$b_1 = 2$$

What is $\hat{y}$ for Zara?

Question 114

Set 4 A

Consider the following example:

Zara's score on $x = 5$

Zara's score on $y = 23$

$$b_0 = 10$$

$$b_1 = 2$$

What is e for Zara?

Question 115

Set 4 A

In the regression equation, what is b_0 ?

In the regression equation, what is b_1 ?

In the regression equation, what is e ?

- A. The expected increase in y given a one unit increase in x
- B. The expected value for y when x equals zero
- C. A person's predicted score on the outcome variable
- D. Observed y minus predicted y

Question 116

Set 4 A

In least squares regression, we calculate an intercept and a slope to minimize something.

Write the formula for the thing that gets minimized.

Question 117

Set 4 A

Use “cov_xy” as the symbol for the covariance between x and y and write the formula for a slope.

Note: If a component of the equation needs both a superscript and a subscript, it does not matter which you type first and which you type second. In class lecture, you were given two ways of writing the formula for a slope, and either way will be counted correct.

Question 118

Set 4 A

In **simple** regression when x and y are **standardized** . . .

a beta will always equal:

an intercept will always equal:

- A. $y - \hat{y}$.
- B. The covariance minus $(y - \hat{y})$.
- C. zero.
- D. the correlation.
- E. the mean of y .

Question 119

Set 4 B

For a **simple regression** with one predictor variable,

Write the full linear regression equation for **variance in a sample**.

Note: The equation should include subscripts.

Question 120

Set 4 B

For a **simple regression** with one predictor variable, there are two equations for sampling error variance in a population. To write the **most important** of these two equations, you would need to include the term “e_1sp.”

Write this linear equation for sampling error variance in a population that includes the term “e_1sp.”

Note: The equation should include subscripts.

If you need to use Greek letters, remember that:

μ = mu

β = beta

ρ = rho

Question 121

Set 4 B

For a **simple regression** with one predictor variable, there are two equations for sampling error variance in a population. To write the **least** important of these two equations, you would need to include the term “e_0sp.”

Write this linear equation for sampling error variance in a population that includes the term “e_0sp.”

Note: The equation should include subscripts.

If you need to use Greek letters, remember that:

μ = mu

β = beta

ρ = rho

Question 122

Set 4 B

As _____ gets steeper, $\Sigma(y - \hat{y})$ gets _____ .

- A. b_0 , larger
- B. b_0 , smaller
- C. b_1 , larger
- D. b_1 , smaller

Question 123

Set 4 B

Use “ss_est” as the symbol for the sum of squared error and write the formula for the standard error of estimate.

Question 124

Set 4 B

$$y_{is} = b_{0s} + b_{1s}x_{is} + e_{is}$$

In the above equation, what is the average deviation of $|e_{is}|$?

- A. The standard deviation
- B. The standard error of estimate
- C. The standard error of beta
- D. Observed y minus predicted y

In the above equation, how is e_{is} calculated?

- A. Cov_{xy}/sd_x^2
- B. $\bar{y} - b_1\bar{x}$
- C. $y - \bar{y}$
- D. $y - \hat{y}$

Question 125

Set 4 B

Correctly spell the term that should be placed in each blank.

In simple regression. . .

$\Sigma(y - \hat{y})^2$ is the sum of squares _____.

$\Sigma(y - \bar{y})^2$ is the sum of squares _____.

$\Sigma(\hat{y} - \bar{y})^2$ is the sum of squares _____.

Question 126

Set 4 B

In simple regression, if $r = 0$, then. . .

the intercept is equal to _____

the sum of squares error is equal to _____

the sum of squares explained is equal to _____

the sum of squares total is equal to _____

**Do NOT say a
term is equal
to itself.**

- A. Zero
- B. Mean of y
- C. Sum of square total
- D. Sum of squares error
- E. Sum of square explained

Question 127

Set 4 B

In simple regression, if $r = 1$, then. . .

the sum of squares error is equal to _____

the sum of squares explained is equal to _____

the sum of squares total is equal to _____

Do NOT say a term is equal to itself.

- A. Zero
- B. Mean of y
- C. Sum of square total
- D. Sum of squares error
- E. Sum of square explained

Question 128

Set 4 B

Imagine that you do a study and find that the total sum of squares is 12 and the error sum of squares is 9.

What is the explained variance sum of squares?

What is the r^2 ?

Question 129

Set 4 C

	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Explained	8	1	8	13.33
Error	42	70	.6	
Total	50	71		

$$b_1 = 1.50$$

$$sd_{beta} = .25$$

If this study were repeated over and over, how much will the beta typically deviate from the true population parameter?

Calculate the value that indicates the standardized e_{1sp} for this study assuming H_0 is true.

Calculate the t -value for the correlation.

Question 130

Set 4 C

	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Explained	30	1	30	3.33
Error	90	10	9	
Total	120	11		

Given the output above, calculate r^2 .

Calculate the value that indicates the average distance that people's predicted scores deviated from their observed scores.

Question 131

Set 4 C

	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Explained	A	B	C	F
Error	D	E	G	
Total	H	I		

Which cell is equal to $\Sigma(y - \bar{y})^2$?

Which cell is equal to $\Sigma(\hat{y} - \bar{y})^2$?

Which cell is equal to $\Sigma(y - \hat{y})^2$?

Which cell is equal to p (number of predictors)?

Which cell is equal to $n - p - 1$?

Which cell is equal to $n - 1$?

Question 132

Set 4 D

You define “fish time” as the number of hours a person spends with pet fish in one week. You do a study using fish time to predict ratings of life wellbeing, and get these results: $b_0 = 8$ and $b_1 = 3$. Lou was a participant in the study, and Lou’s fish time was 4 hours.

What is Lou’s predicted life wellbeing?

If Lou’s actual life wellbeing score was 25, what is Lou’s life wellbeing with fish time extracted out of it?

Question 133

Set 4 D

You have scores on variable x and variable y for a sample of people. You extract x out of y to create variable e . If the correlation between x and y is **zero**, then. . .

the correlation between x and e is equal to _____

$y - \hat{y}$ is equal to _____

$\bar{e}$ is equal to _____

sd_e is equal to _____

- A. Zero
- B. One
- C. $y - \bar{y}$
- D. sd_x
- E. sd_y
- F. $\bar{y}$

Question 134

Set 4 D

You have scores on variable x and variable y for a sample of people. You extract x out of y to create variable e . If there is a **perfect** correlation between x and y , then. . .

the correlation between x and e is equal to _____

$y - \hat{y}$ is equal to _____

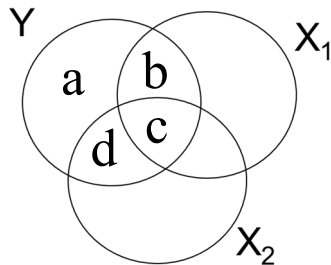
$\bar{e}$ is equal to _____

sd_e is equal to _____

- A. Zero
- B. One
- C. $y - \bar{y}$
- D. sd_x
- E. sd_y
- F. $\bar{y}$

Question 135

Set 5 A

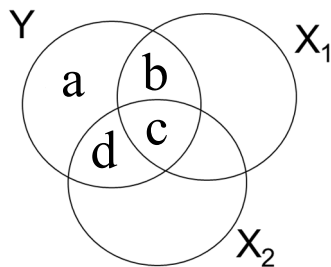


Using the letters from the Venn diagram above, type the formula that would be equal to R^2 .

Using the letters from the Venn diagram above, type the formula that would be equal to r^2 for the association between x_1 and y .

Question 136

Set 5 A



Possible answers

Area b
Area $b + c$
Area $b + c + d$
a correlation
an R^2
a standardized beta
a t -test
an F -test

If you want to know the effect of x_1 ignoring the other x variables:

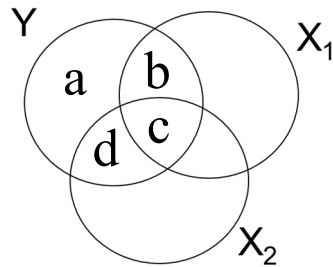
In the Venn diagram, this pertains to _____.

The effect size is given by _____.

The significance of the effect is tested with _____.

Question 137

Set 5 A



Possible answers

Area b
Area b + c
Area b + c + d
a correlation
an R^2
a standardized beta
a t -test
an F -test

If you want to know the effect of all the x variables taken together:

In the Venn diagram, this pertains to ____ .

The effect size is given by ____ .

The significance of the effect is tested with ____ .

Question 138

Set 5 A

For which of the following statistics is the expected value the **same** as the population parameter?

- A. R
- B. R^2
- C. r
- D. r^2

Question 139

Set 5 A

Below are some results from a study in which a researcher was using multiple regression to predict outcome y .

	SS	df	MS	F
Explained	8	3	2.67	4.09
Error	30	46	0.65	
Total	20	49		

$$r = .28$$

$$\sqrt{\frac{1 - r^2}{n - 2}} = .14$$

How many predictor variables were used in the regression equation?

Question 140

Set 5 A

Below are some results from a study in which a researcher was using multiple regression to predict outcome y .

	SS	df	MS	F
Explained	8	3	2.67	4.09
Error	30	46	0.65	
Total	20	49		

$$r = .28$$

$$\sqrt{\frac{1 - r^2}{n - 2}} = .14$$

Based on this output, what is a value for an **inferential** statistic that could be used to test the significance of the effect of one x variable **ignoring** the other x variables?

Question 141

Set 5 A

Below are some results from a study in which a researcher was using multiple regression to predict outcome y .

	SS	df	MS	F
Explained	8	3	2.67	4.09
Error	30	46	0.65	
Total	20	49		

$$r = .28$$

$$\sqrt{\frac{1 - r^2}{n - 2}} = .14$$

Based on this output, what is a value for an **inferential** statistic that could be used to test the significance of the effect of **all** the x variables taken together?

Question 142

Set 5 A

Below are some results from a study in which a researcher was using multiple regression to predict outcome y .

	SS	df	MS	F
Explained	8	3	2.67	4.09
Error	30	46	0.65	
Total	20	49		

$$r = .28$$

$$\sqrt{\frac{1 - r^2}{n - 2}} = .14$$

Based on this output, what is a value for a **descriptive** statistic that could be used to give an unbiased **effect size** for the effect of one x variable **ignoring** the other x variables?

Question 143

Set 5 A

Below are some results from a study in which a researcher was using multiple regression to predict outcome y .

	SS	df	MS	F
Explained	8	3	2.67	4.09
Error	30	46	0.65	
Total	20	49		

$$r = .28$$

$$\sqrt{\frac{1 - r^2}{n - 2}} = .14$$

Based on this output, what is a value for a **descriptive** statistic that could be used to give the **effect size** for the percent of variance explained by **all** the x variables taken together.

Question 144

Set 5 A

Below are some results from a study in which a researcher was using multiple regression to predict outcome y .

	SS	df	MS	F
Explained	8	3	2.67	4.09
Error	30	46	0.65	
Total	20	49		

$$r = .28$$

$$\sqrt{\frac{1 - r^2}{n - 2}} = .14$$

Based on this output, what is the correlation between observed y and predicted y ?

Question 145

Set 5 B

Fill in each blank with a numerical value or with a correctly spelled component from the equation (with its subscript, if relevant).

$$y = b_0 + b_1x_1 + b_2x_2 + e$$

In the regression equation above, b_1 is equal to the predicted increase in _____ for a _____ unit increase in _____ when _____ is equal to _____.

Question 146

Set 5 B

$$y = b_0 + b_1x_1 + b_2x_2 + e$$

Class lecture described a method that could be used for calculating b_1 from the equation above. One of the steps in this method involved calculating a residual score for y .

Write the regression equation, with subscripts, that would be used to calculate the needed residual score for y .

Question 147

Set 5 B

$$y = b_0 + b_1x_1 + b_2x_2 + e$$

Class lecture described a method that could be used for calculating b_1 from the equation above. One of the steps in this method involved calculating a residual score for x_1 .

Write the regression equation, with subscripts, that would be used to calculate the needed residual score for x_1 (that is, the residual score for x_1).

Question 148

Set 5 B

$$y = b_0 + b_1x_1 + b_2x_2 + e$$

Class lecture described a method that could be used for calculating b_1 from the equation above. The final step in this method involved estimating a regression equation that included two residual scores and that produced a value for b_1 that equaled b_1 from the equation above.

Write this regression equation, with subscripts, that includes two residual scores and produces the value for b_1 in the equation above.

Question 149

Set 5 B

Fill in the bank with a correctly spelled component from the equation (with its subscript, if relevant).

Equation 1 $y = b_0 + b_1x_1 + b_2x_2 + e$

Equation 2 $e_{y.x1} = b_0 + b_2e_{x2.x1} + e$

b_2 from Equation 1 is exactly equal to _____ from Equation 2.

(Note, this questions is asking about b_2 and not b_1 .)

Question 150

Set 5 B

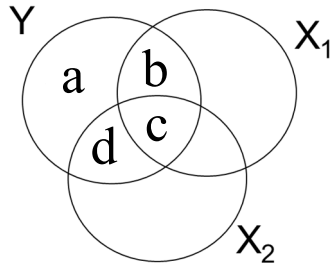
$$y = 5 + 4x_1 + 6x_2 + e$$

According to the equation above, y is expected to increase by 4 units when x_1 increases by 1 units and x_2 _____ .

- A. is 6.
- B. is 0.
- C. is 5.
- D. increases by 6 units

Question 151

Set 5 C



Possible answers

Area b
Area b + c
Area b + c + d
a correlation
an R^2
a standardized beta
a t -test
an F -test

If you want to know the effect of x_1 after controlling for x_2 :

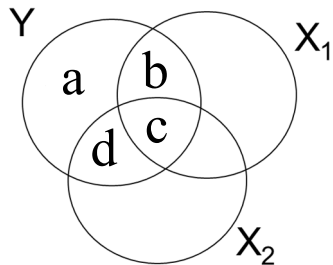
In the Venn diagram, this pertains to ____ .

The effect size is given by ____ .

The significance of the effect is tested with ____ .

Question 152

Set 5 C



Using the letters from the Venn diagram above,
type the formula that would be equal to a
squared partial correlation between x_1 and y .

Question 153

Set 5 C

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + e$$

You could calculate the **unstandardized beta weight** for x_1 above by using a **simple** regression equation in which [A] is used to predict [B].

What belongs in space [A]?

- A. the raw score for x_1
- B. the z-score for x_1
- C. the residual for x_1
- D. the residual of the z-score for x_1 (meaning that x_1 is a z-score)
- E. the z-score of the residual for x_1 (meaning that the residual is a z-score)

What belongs in space [B]?

- A. the raw score for y
- B. the z-score for y
- C. the residual for y
- D. the residual of the z-score for y (meaning that y is a z-score)
- E. the z-score of the residual for y (meaning that the residual is a z-score)

Question 154

Set 5 C

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + e$$

You could calculate the **standardized beta weight** for x_1 above by using a **simple** regression equation in which [A] is used to predict [B].

What belongs in space A?

- A. the raw score for x_1
- B. the z-score for x_1
- C. the residual for x_1
- D. the residual of the z-score for x_1 (meaning that x_1 is a z-score)
- E. the z-score of the residual for x_1 (meaning that the residual is a z-score)

What belongs in space B?

- A. the raw score for y
- B. the z-score for y
- C. the residual for y
- D. the residual of the z-score for y (meaning that y is a z-score)
- E. the z-score of the residual for y (meaning that the residual is a z-score)

Question 155

Set 5 C

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + e$$

You could calculate the **partial correlation** for x_1 above by using a **simple** regression equation in which [A] is used to predict [B].

What belongs in space A?

- A. the raw score for x_1
- B. the z-score for x_1
- C. the residual for x_1
- D. the residual of the z-score for x_1 (meaning that x_1 is a z-score)
- E. the z-score of the residual for x_1 (meaning that the residual is a z-score)

What belongs in space B?

- A. the raw score for y
- B. the z-score for y
- C. the residual for y
- D. the residual of the z-score for y (meaning that y is a z-score)
- E. the z-score of the residual for y (meaning that the residual is a z-score)

Question 156

Set 5 C

You calculate a statistic and get a value of .2 for x_1 and a value of .8 for x_2 . The effect for x_1 might be larger than the effect for x_2 if the value you calculated is a

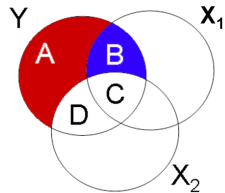
- A. beta
- B. standardized beta
- C. partial correlation
- D. none of the above

Which of the following is true about a standardized beta?

- A. It cannot be greater than 1 and cannot be less than -1.
- B. Residuals are extracted from x and y and then the residuals are standardized.
- C. Both x and y are standardized prior to extracting residuals.
- D. It will have a t -value that is smaller than the t -value for an unstandardized beta.

Question 157

Set 5 C



Which of the following is equal to area **B** divided by area **A + B**?

- A. t -value for beta
- B. squared beta
- C. squared standardized beta
- D. squared partial correlation

What descriptive statistic would you calculate if you want to know the size of the standardized association between x_1 and y after controlling for other variables?

- A. t -value
- B. standardized beta
- C. r
- D. R^2

Question 158

Set 5 C

Thing 1: $\frac{b}{sd_{beta}}$

Thing 2: the t -value for the partial correlation

Thing 3: the t -value for $b \frac{sd_x}{sd_y}$

Which of the three things are equal to each other?

- A. None of them are equal
- B. Thing 1 and Thing 2
- C. Thing 1 and Thing 3
- D. Thing 2 and Thing 3
- E. All three things are equal

Question 159

Set 5 C

$$\sqrt{\frac{t^2}{t^2 + df}}$$

What does this formula do?

- A. Turns a t -value into a correlation effect size
- B. Turns a t -value into an inferential statistic.
- C. Turns a t -value into a standardized beta.
- D. Turns a t -value into $\frac{b}{sd_{beta}}$
- E. Turns a t -value into a value that could be larger than one.

Question 160

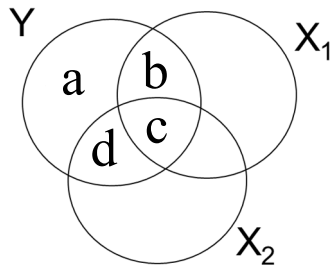
Set 5 D

The key difference between a partial correlation and a semipartial correlation has to do with:

- A. how the residuals for the x variables are defined.
- B. whether variables are standardized before or after extracting residuals.
- C. how the variance for y is defined.
- D. Whether it is possible to compare relative unique contributions of each x variable.

Question 161

Set 5 D



Using the letters from the Venn diagram above, type the formula that would be equal to a squared semi-partial correlation between x_1 and y .

Question 162

Set 5 D

Fill in the blank with a correctly spelled term:

ΔR^2 is equal to R^2 full minus R^2 _____.

Question 163

Set 5 D

In the first step of an analysis, income and education were used to predict happiness. Which of the following are potential second steps that would allow the computation of ΔR^2 ? Select all examples that are potential second steps.

- A. Income and education are used to predict **anger**.
- B. Income, education, and **trust** are used to predict happiness.
- C. **Trust** and **agreeableness** are used to predict happiness.
- D. Income, education, **trust** and **agreeableness** are used to predict happiness.

Question 164

Set 5 D

The table below displays correlations between four variables from a sample of 100 people. If you used all three x variables to predict y , which variables are likely to have significant beta weights?

- A. None of them
- B. x_2
- C. x_1 and x_3
- D. All of them

	x_1	x_2	x_3	y
x_1	1			
x_2	.31	1		
x_3	.87	.28	1	
y	.40	.44	.47	1

Question 165

Set 5 D

Correlations

	x_1	x_2	y
x_1	1		
x_2	.31	1	
y	.60	.54	1

$$\text{Equation 1: } y = b_0 + b_1x_1 + e$$

In Equation 1, the p -value for b_1 is .02

$$\text{Equation 2: } y = b_0 + b_1x_1 + b_2x_2 + e$$

Given these results, what best describes how results for Equation 2 are **likely** to compare with results for Equation 1?

- A. In Equation 2, R^2 will increase, and b_1 will be larger.
- B. In Equation 2, R^2 will increase, and b_1 will be smaller.
- C. In Equation 2, R^2 will increase, but only a trivial amount and, and b_1 will become non-significant.
- D. In Equation 2, R^2 will decrease, and b_1 will be larger.
- E. In Equation 2, R^2 will decrease, and b_1 will be smaller.

Question 166

Set 5 D

Correlations

	x_1	x_2	y
x_1	1		
x_2	.92	1	
y	.48	.50	1

$$\text{Equation 1: } y = b_0 + b_1x_1 + e$$

In Equation 1, the p -value for b_1 is .02

$$\text{Equation 2: } y = b_0 + b_1x_1 + b_2x_2 + e$$

Given these results, what best describes how results for Equation 2 are **likely** to compare with results for Equation 1?

- A. In Equation 2, R^2 will increase more than a trivial amount, and b_1 will be smaller.
- B. In Equation 2, R^2 will increase, but only a trivial amount and, and b_1 will become non-significant.
- C. In Equation 2, R^2 will decrease, and b_1 will be larger.
- D. In Equation 2, R^2 will decrease, and b_1 will become non-significant.

Question 167

Set 5 D

Pick the answer that would make this an example of a **suppressor effect**.

The correlation between x_1 and y was .24 and significant. After controlling for x_2 , the partial correlation between x_1 and y was _____ .

- A. Larger
- B. About the same
- C. Smaller but still significant
- D. Not significant
- E. Different from the standardized beta

Question 168

Set 6 A

$$\bar{y}_1 - \bar{y}_0 = ?$$

The equation depicted above provides an example of _____ . (Select all the correct answers.)

- A. A contrast
- B. A focused question
- C. An omnibus question

Question 169

Set 6 A

The values below include a grand mean and three groups means from a study contrasting three groups. Ollie Otter is in group 1. What is Ollie Otter's predicted value on the outcome variable y ?

$$\bar{x}_{grand} = 13 \quad \bar{x}_1 = 9 \quad \bar{x}_2 = 11 \quad \bar{x}_3 = 16$$

Question 170

Set 6 A

Fill in each blank with a single word that is correctly spelled.

When using ANOVA to test for differences between groups. . .

the sum of squares **explained** is typically called the sum of squares _____ groups,

and the sum of squared **error** is typically called the sum of squares _____ groups.

Question 171

Set 6 A

When using ANOVA to test for differences between groups the formula for the sum of squares for **explained variance** can be written as:

$$SS_{treatment} = \Sigma \text{_____} .$$

Write the remaining part of the formula (just the part of the formula that appears **after** the Σ sign).

Question 172

Set 6 A

When using ANOVA to test for differences between groups the formula for the sum of squares for **error variance** can be written as:

$$SS_{error} = \Sigma \text{_____} .$$

Write the remaining part of the formula (just the part of the formula that appears **after** the Σ sign).

Question 173

Set 6 A

When using ANOVA to test for differences between groups the formula for the sum of squares for **total variance** can be written as:

$$SS_{total} = \Sigma \text{_____} .$$

Write the remaining part of the formula (just the part of the formula that appears **after** the Σ sign).

Question 174

Set 6 A

Assume you are using ANOVA to test for differences between groups. If you have 5 groups, how many contrasts are free to vary?

Question 175

Set 6 A

Assume you are using ANOVA to test for differences between groups and you get the following results.

	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Explained	30	2	15	1.67
Error	90	10	9	
Total	120	12		

What is standard deviation for y ?

What is the pooled within-group standard deviation for y ?

What is η^2 ?

How many groups were used in the analysis?

Question 176

Set 6 A

sd_{est} is the same thing as:

- A. η^2
- B. ω^2
- C. The pooled within group sd
- D. The total, grand sd

Question 177

Set 6 A

η^2 will be _____ R^2 and η^2 will be _____ ω^2 .

- A. larger than
- B. the same as
- C. smaller than

- A. larger than
- B. the same as
- C. smaller than

Question 178

Set 6 B

Correctly spell the term that belongs in the blank.

The process of creating predictor variables with values that are determined by group membership is called _____ .

Question 179

Set 6 B

Assume that x is group membership (coded such that 1 = treatment and 0 = control), and that a regression analysis with this variable produces the following results:

$$\hat{y} = 12 + 3x$$

The t -value for the beta in this equation (with 75 df) is 4.

$$\sqrt{\frac{4^2}{4^2 + 75}} = .42$$

What is $\bar{y}_1 - \bar{y}_0$?

What is $\bar{y}_0$?

What is the standardized beta?

What is $\frac{cov_{xy}}{sd_x sd_y}$?

What is η^2 ?

What is R^2 ?

Question 180

Set 6 B

Assume that x is group membership (coded such that 1 = treatment and 0 = control), and that a regression analysis with this variable produces the following results:

$$\hat{y} = 12 + 3x$$

The t -value for the beta in this equation (with 75 df) is 5.

$$\sqrt{\frac{5^2}{5^2 + 75}} = .5$$

What is the mean for the treatment group?

What is the predicted value for someone in the control group?

Question 181

Set 6 B

Assume that x is group membership (coded such that 1 = treatment and 0 = control), and that a regression analysis with this variable produces the following results:

$$\hat{y} = 12 + 3x$$

The t -value for the beta in this equation (with 75 df) is 5.

$$\sqrt{\frac{5^2}{5^2 + 75}} = .5$$

What is the difference between means divided by the standard error of the difference between means?

What is the F value? (Hint, this will require squaring something).

Question 182

Set 6 C

If a researcher wants to create orthogonal contrasts for a study comparing 5 groups, how many contrasts does the researcher need to create?

Question 183

Set 6 C

	Group 1	Group 2	Group 3
X1	-.66	.33	.33
X2	0	-.5	.5

If the contrasts depicted in the code assignment chart above are entered into a regression equation, the equation for someone in **Group 3** would be: $\hat{y} = \underline{\hspace{2cm}}$.

Write the remaining part of the formula (just the part of the formula that appears **after** the “=” sign) and use **numerical values** instead of the letter “x.”

Question 184

Set 6 C

x_1	.33	E	F
x_2	G	H	-.5

Orthogonal contrasts are being used to create contrasts for an analysis with three groups. The code assignment chart depicted above has four missing values (labeled E, F, G, H).

Following procedures discussed in class lecture, figure out what the missing values need to be and then write the complete code assignment chart with all the correct values.

Question 185

Set 6 C

Assume you are creating two orthogonal contrasts. In the table below, list values you could use for the first contrast so that people in group 3 are given a negative value, and so that the contrast compares people in group 3 with people in groups 1 and 2.

	Group 1	Group 2	Group 3
x_1	A	B	C

Question 186

Set 6 C

Assume you are creating two orthogonal contrasts. In the table below, list values you could use for the first contrast so that people in group 3 are given a negative value, and so that the contrast compares people in group 3 with people in group 1 (and group 2 is not included in the comparison).

	Group 1	Group 2	Group 3
x_1	A	B	C

Question 187

Set 6 C

	Group 1	Group 2	Group 3
X_1	-.66	.33	.33
X_2	0	-.5	.5

Mean scores on the outcome variable are 4, 8, and 12 for groups 1, 2, and 3, respectively.

Assume the contrasts listed above are entered into a regression equation:

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

Given the results listed above:

What is b_0 ?

What is b_1 ?

What is b_2 ?

Question 188

Set 6 C

	Group E	Group F	Group G
x_1	-.66	.33	.33
x_2	0	-.5	.5

The code assignment chart depicted above was used in an analysis that produced the following results:

$$\hat{y} = b_0 + 6x_1 + 2x_2 \quad \bar{y}_{\text{Group F}} = 9$$

What is the mean for group G?

What is the mean for group E?

What is b_0 ?

Question 189

Set 6 C

Orthogonal contrasts are being used to create contrasts for an analysis with three groups. What is the name of the **descriptive statistic** that provides an effect size for the difference between Group 2 and Group 3?

- A. F
- B. η^2
- C. t
- D. d

Question 190

Set 6 C

Orthogonal contrasts are being used to create contrasts for an analysis with three groups. What is the name of the **inferential statistic** that tests the significance of the difference between Group 2 and Group 3?

- A. F
- B. η^2
- C. t
- D. d

Question 191

Set 6 C

	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Explained	30	2	15	1.67
Error	90	10	9	
Total	120	12		

	Group 1	Group 2	Group 3
X_1	-.66	.33	.33
X_2	0	-.5	.5

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

$$b_0 = 15 \quad b_1 = 2.4 \quad b_2 = 1.5$$

Based on the equation, the orthogonal contrasts, and the results listed above:

What is the d -statistic for the contrast between Group 1 and the other two groups?

What is the d -statistic for the contrast between Group 2 and Group 3?

Question 192

Set 6 D

You do a study with many groups and gets these ANOVA results.

	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Explained	84	4	21	1.31
Error	304	19	16	
Total	388	23		

If you wanted to do a t -test for a contrast between just Group 1 and Group 2, you would need the formula below. However, because there are more than 2 groups in this study, you would need to replace sd_{pooled}^2 with a value that can be calculated using the results above. What is this value?

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{sd_{pooled}^2}{n_1} + \frac{sd_{pooled}^2}{n_2}}$$

Question 193

Set 6 D

You do a study with many groups and gets these ANOVA results.

	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Explained	84	4	21	1.31
Error	304	19	16	
Total	388	23		

If you wanted to do a t -test for a contrast between just Group 1 and Group 2, what are the appropriate degrees of freedom to use for this t -test?

Question 194

Set 6 D

A researcher conducted 20 different t -tests, and in every case the null hypothesis was true. What is the best guess for how many of those t -tests were significant?

Question 195

Set 6 D

Dug the dog used t -tests to test contrasts between pairs of groups in a study with 4 groups. The results for the contrast between Group 1 and Group 2 are: $t(230) = 2.5, p = .02$.

If Dug the dog made all the possible contrasts between group pairs, what is the p value for this contrast after a Bonferroni correction?

If Dug the dog tested only a set of planned comparisons following standard procedures for orthogonal contrasts, what is the p value for this contrast after a Bonferroni correction?

Question 196

Set 6 D

Keeping in mind the fact that the most stringent procedures for protecting against family-wise error will produce the largest p -values, fill in the blanks to reflect the most likely outcome.

A researcher used three different methods to protect against family-wise error and got p -values of .07, .04, and .02. It is most likely that:

_____ produced a p -value of .07

_____ produced a p -value of .04

_____ produced a p -value of .02

- A. Fisher's Least Significant Difference procedure
- B. A Bonferroni correction
- C. A Tukey Honestly Significant Difference test

Question 197

Set 6 D

Post hoc analysis is used to test all the possible paired group contrasts. What is the name of the inferential statistic that can be used to determine if the difference between Group 2 and Group 3 is significant?

- A. η^2
- B. F
- C. q
- D. d